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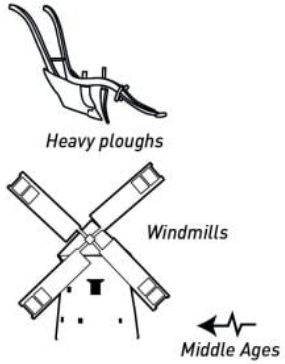
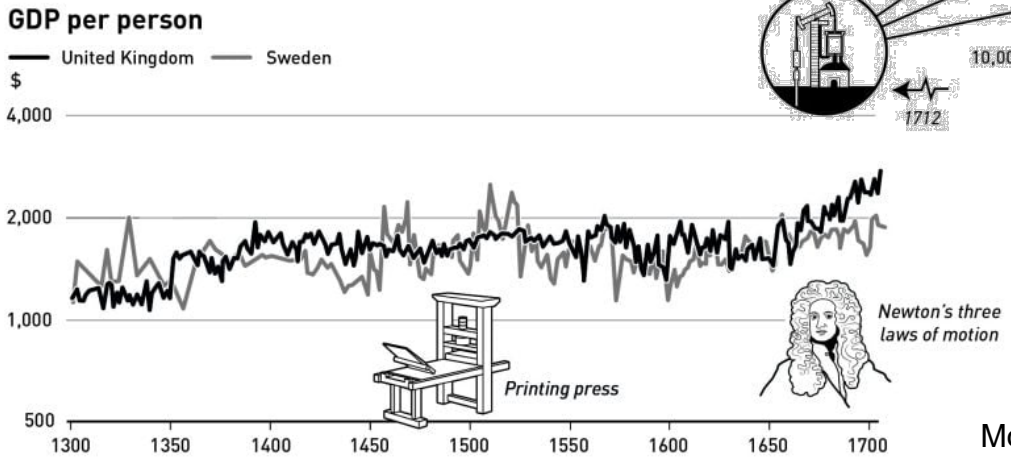
Principles of Macroeconomics

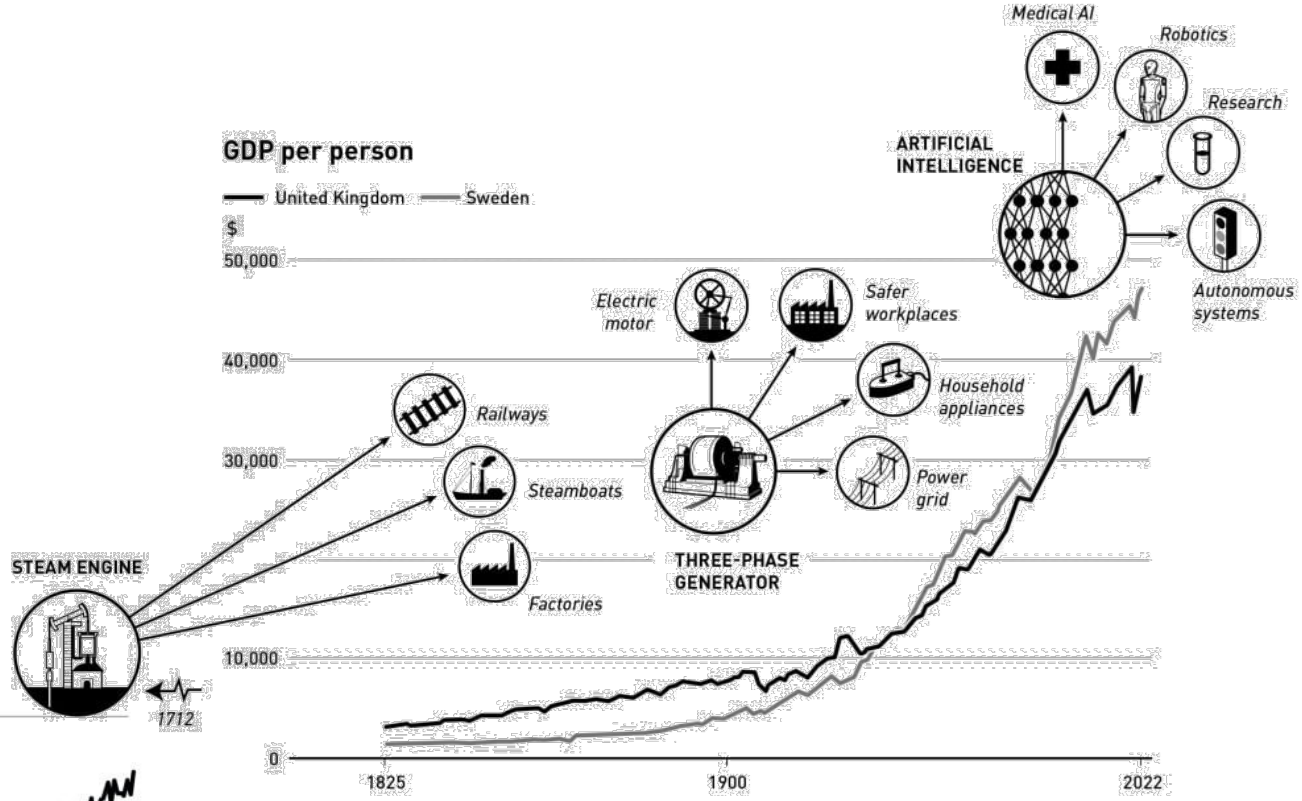
Lecture 5: Saving, investment and the financial system

Prof. Dr. Jan-Egbert Sturm

2025-Nobel Prize: Joel Mokyr / Philippe Aghion / Peter Howitt
– *for studies on how new technology can drive sustained growth*







More information: https://www.nobelprize.org/prizes/economic-sciences/2025/

2025-Nobel Prize: Joel Mokyr / Philippe Aghion / Peter Howitt – *for studies on how new technology can drive sustained growth*

- The laureates show that sustained growth arises when new ideas continuously replace old ones through innovation and creative destruction
- They emphasize the interplay between scientific knowledge, practical application, and institutions that reward openness to new ideas
- Slowing growth in advanced economies may stem from rising market concentration that weakens innovation incentives
- Innovation creates adjustment costs requiring policies that protect workers rather than specific jobs
- New technologies such as AI can boost growth but also intensify social and environmental risks
- Sustained progress depends on openness, competition, and well-designed policies that foster innovation while managing its side effects

- More information: https://www.nobelprize.org/prizes/economic-sciences/2025/

Schedule

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Principles of Macroeconomics

Lecture 4: Unemployment and the Labour Market

Prof. Dr. Jan-Egbert Sturm

(1) How Is Unemployment Measured? Definitions

- To measure unemployment national authorities aim at splitting the entire population into three groups:
  1. ***Employed***
     - A person with a job is considered as employed.
  2. ***Unemployed***
     - A person is considered as unemployed, if that person does not have a job and(!) is able and available (willing) to work at current wage rates.
  3. ***Not in the Labour Force***
     - People who do not have a job and are either not able or not available to work

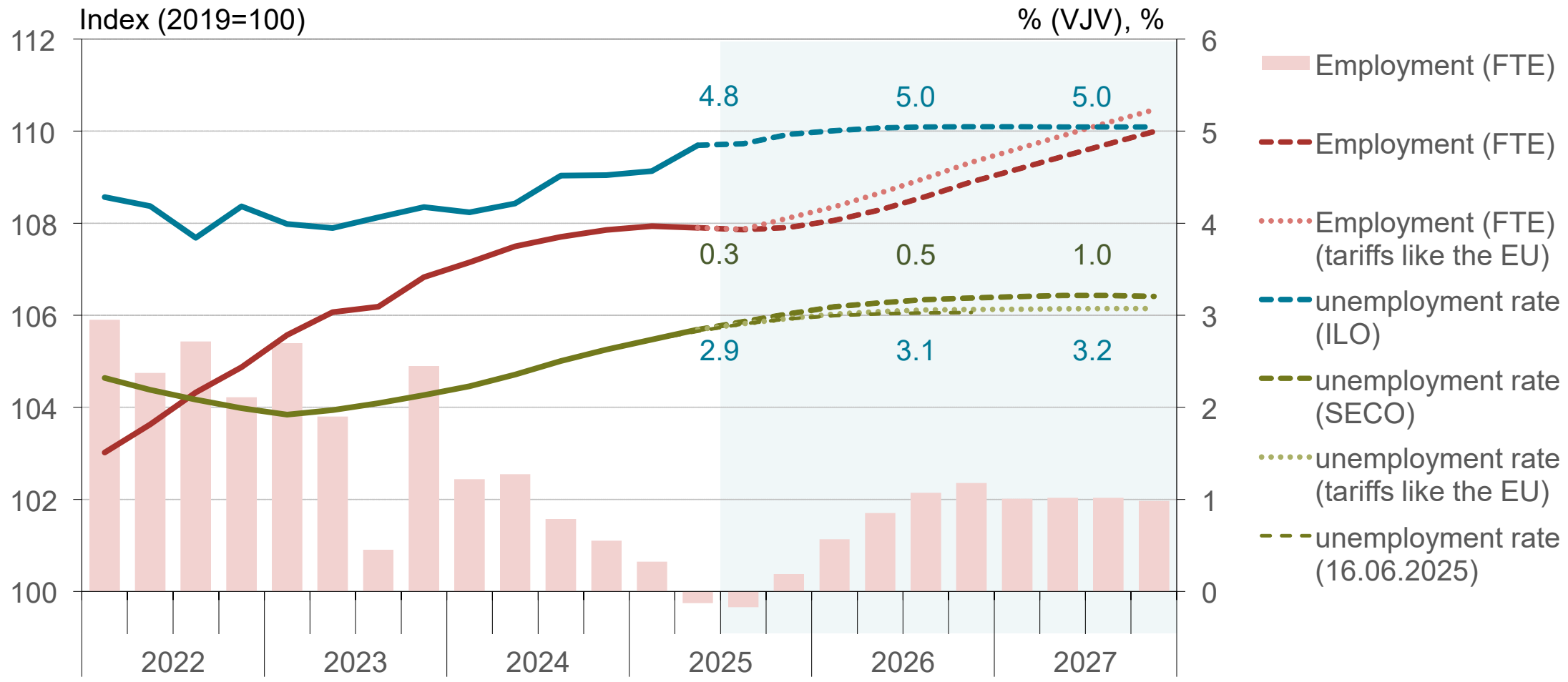
- The ***Labour Force*** is the total number of (employed and unemployed) workers:
  Labour Force = Employed + Unemployed

(1) How Is Unemployment Measured?

1. The ***labour force participation rate*** is the percentage of the adult population that is in the labour force
   - Labour Force Participation Rate = (Number of persons in the labour force / Adult population) x 100
     - Where any person aged 16 or older counts to the adult population (when using the standardised definition)

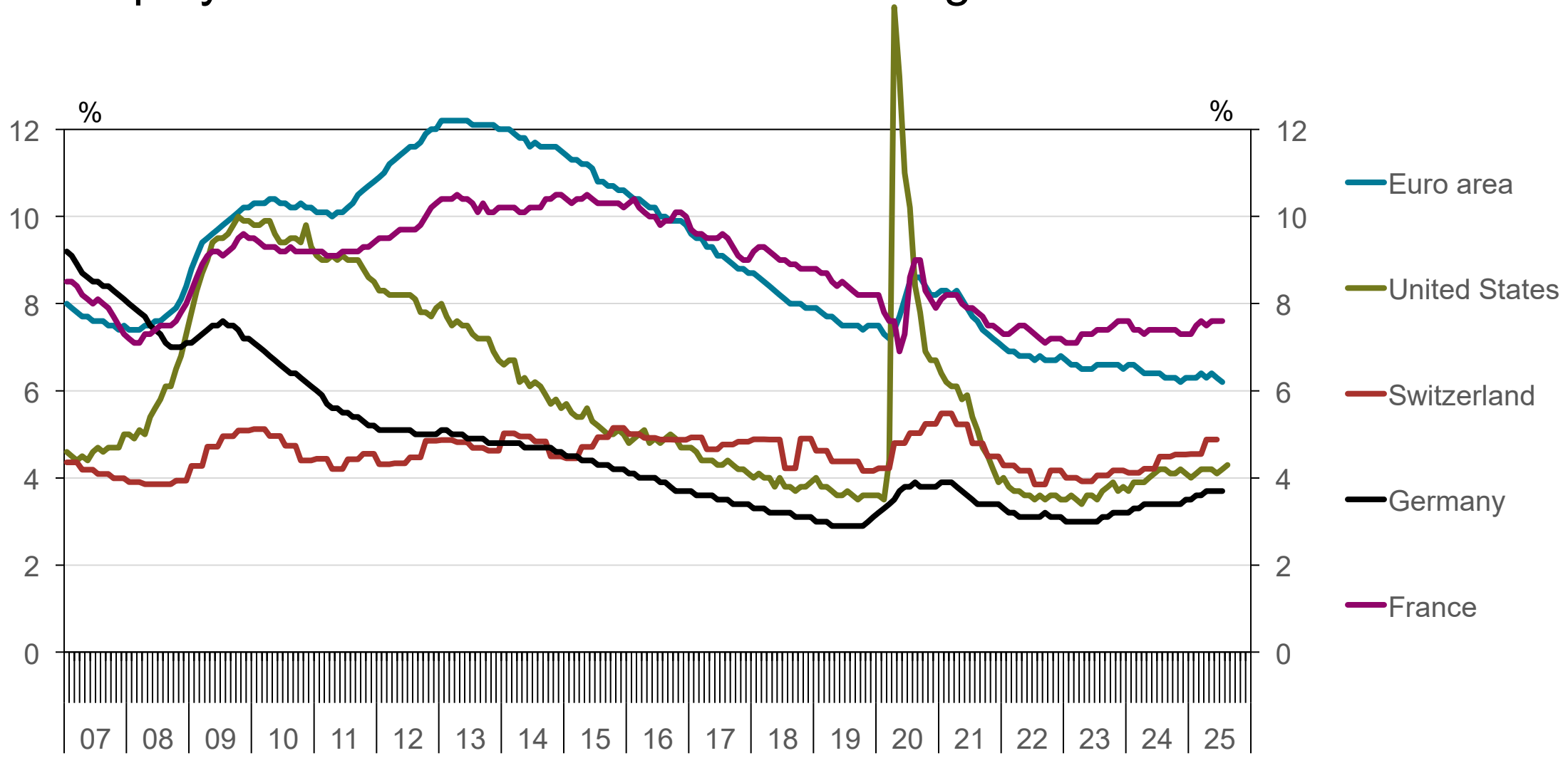
2. The ***unemployment rate*** is calculated as the percentage of the labour force that is unemployed
   - Unemployment rate = (Number of unemployed / Labour force) x 100
   - In practise measured in two ways
     - The claimant count
       - In Switzerland done by SECO (“Arbeitslosenquote”)
     - A monthly survey of households
       - In Switzerland done by the SFSO (ILO based figures) (“Erwerbslosenquote”)

Labour market developments in Switzerland



Quellen: SECO, BFS, KOF (24.09.2025)

Unemployment rate in western countries/regions



Source: OECD, Eurostat

(2) Does the Unemployment Rate Measure What We Want It To?

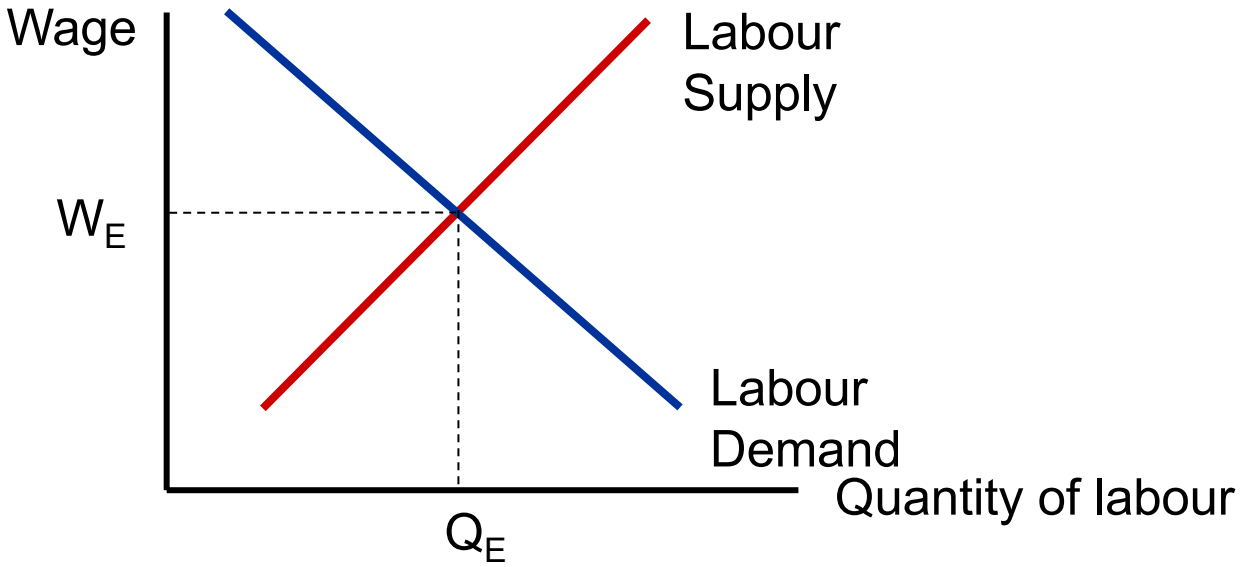
- Problem: It is difficult to distinguish between a person who is unemployed and a person who is not in the labour force
  - ***Discouraged workers***, people who actually would like to work but are discouraged and have given up looking for jobs after unsuccessful searches. Consequently, these don't show up in unemployment statistics anymore and are – with the current measures – considered not to be in the labour force
  - Other people may claim to be unemployed in order to receive financial assistance, even though they aren't looking for work or may even have (unregistered) working opportunities
- These instances may confound our measures

(3) How Long Are the Unemployed without Work?

- In judging how serious the unemployment problem is, we need to consider the duration of unemployment
  - Economists have devoted much energy to study data on unemployment duration and come up with an important, subtle and seemingly contradictory finding:
    - «Most spells of unemployment are short,
      and most unemployment observed at any given time is long term.»
  - Hence, most people who become unemployed will soon find jobs.
    Yet most of the economy's unemployment problem is attributable to relatively few workers who are jobless for long periods of time
    - ***Hysteresis*** refers to the effect is that the longer people are without work the less likely they are to be hired by firms

(4) Why Are There Always Some People Unemployed?

- In an ideal labour market, wages would adjust to balance the supply and demand for labour, ensuring that all workers would be fully employed
- In equilibrium there is no unemployment, because everybody who is willing to work at the given market wage is able to do so (market clears)
  - Everybody who is not working does not want to and will not look for work at current wage levels

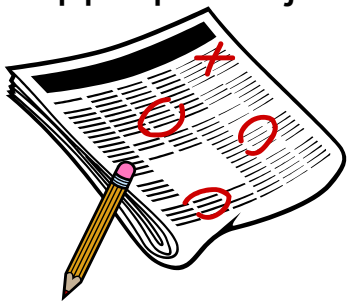


(4) Why Are There Always Some People Unemployed?

- Two different types of unemployment are responsible for long-term unemployment:
  - ***Frictional unemployment***
    - Unemployment that results from the time that it takes to match workers with jobs that are best suit their tastes and skills
    - This type is often thought to explain relatively short spells of unemployment
  - ***Structural unemployment***
    - Unemployment that results because the number of jobs available in some labour markets is insufficient to provide a job for everyone who wants one (labour supply exceeds labour demand)
    - It is often thought to explain longer spells of unemployment.
- We examine four different ways to explain unemployment in the long-run:
  - Frictional unemployment due to (A) job search
  - Structural unemployment due to (B) minimum wage laws
    (C) unions
    (D) efficiency wages

(4A) Job Search

- One reason why economies always experience some unemployment is job search.
  - ***Job search*** is the process by which workers find appropriate jobs given their tastes and skills
  - It results from the fact that it takes time for qualified individuals to be matched with appropriate jobs
  - This unemployment is different from the other types of unemployment:
    - It is not caused by a wage rate higher than equilibrium (as in the case of structural unemployment)
    - It is caused by the time spent searching for the “right” job

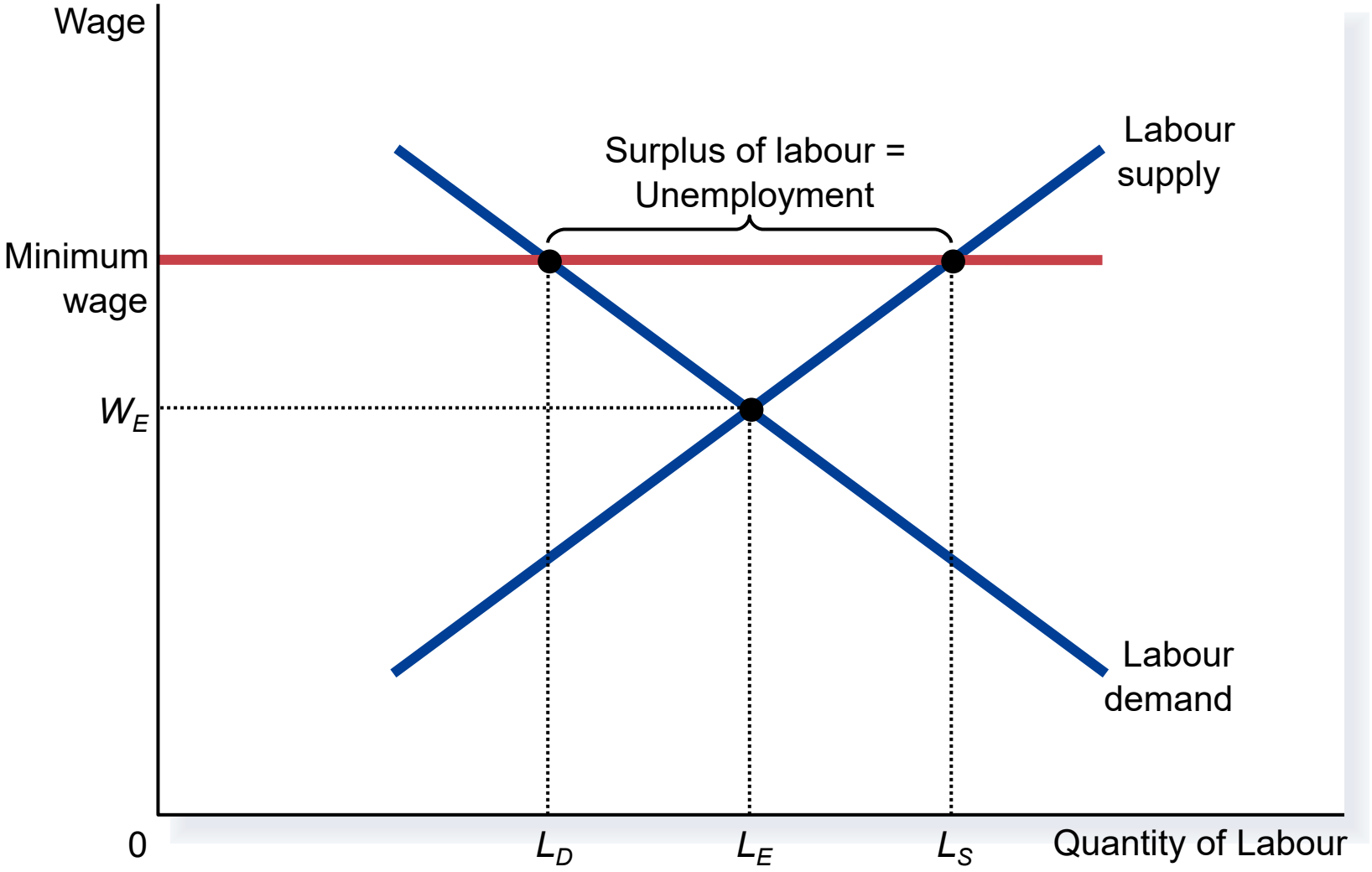


- Job search related frictional unemployment is inevitable because the economy changes continuously
  - Workers leave their jobs because they realize that the jobs are no a good match
  - If consumers start prefer Brand A over Brand B, then B lays workers of while A hirers workers
  - Technological and changes in preferences lead to sectoral shifts
    - If the oil price falls, oil extracting firms cut back their production and employment. At the same time, stimulated car sales cause car manufacturers to increase their production and employment
  - …

(4A) Job Search and Public Policy

- Government programs can affect the time it takes unemployed workers to find new jobs and firms to find the right people
- These programs include the following:
  - Government-run ***employment agencies*** give out information about job vacancies in order to match workers and jobs more quickly
  - Public ***training programs*** aim to ease the transition of workers from declining to growing industries and to help disadvantaged groups escape poverty.
  - ***Unemployment insurance*** is a government program that partially protects workers' incomes when they become unemployed.
- There are advocates and critics of these policies:
  - Advocates believe the economy to operate more efficiently by keeping labour force more fully employed and by reducing the economy's inherent inequities
  - Critics question government intervention on grounds of efficiency and condemn that it may shrink the incentives to become re-employed

(4B) Minimum-Wage Laws



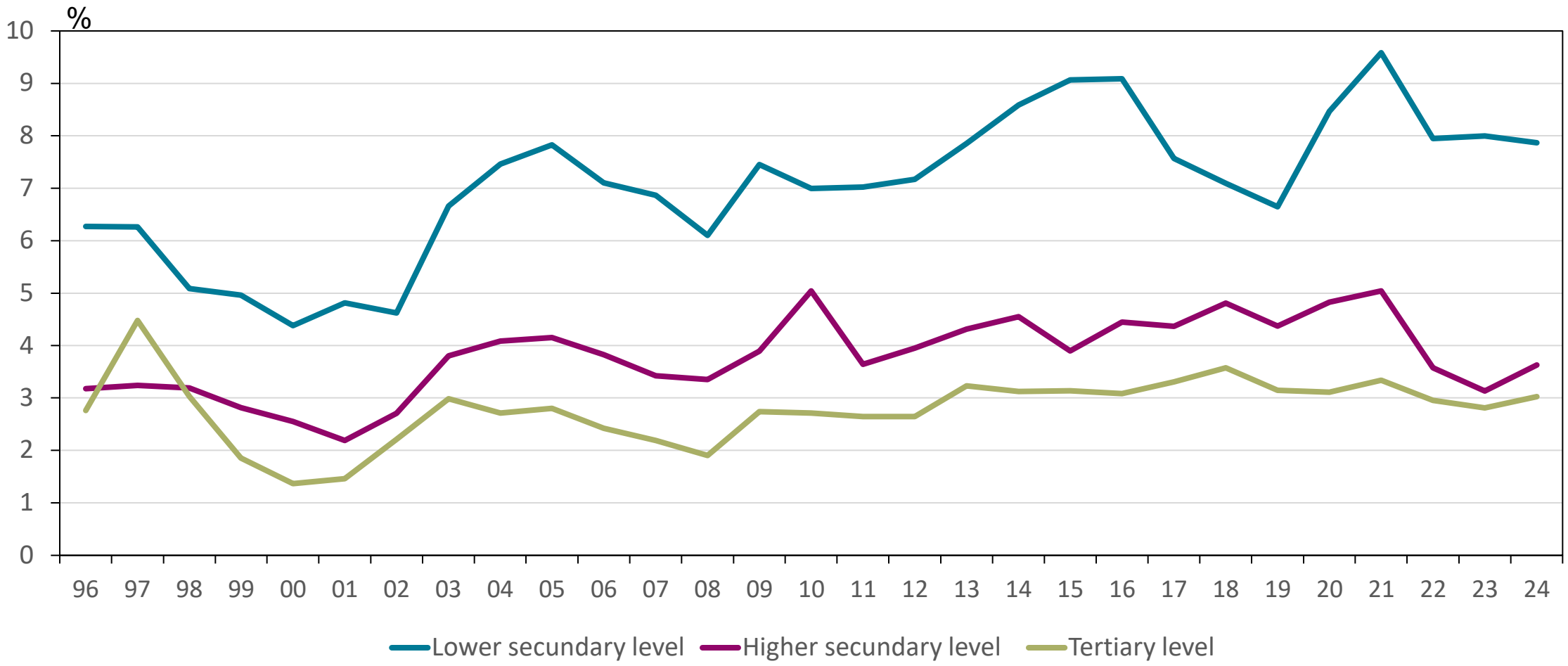
- When the minimum wage is set above the level that balances supply and demand (equilibrium level), it creates unemployment
- Social welfare benefits function like a minimum wage

Case Study: Minimum wage increase in the U.S. in 1996

Unemployment rates, before & after		
	3rd Q 1996	1st Q 1997
Teenagers	16.6%	17.0%
Single mothers	8.5%	9.1%
All workers	5.3%	5.3%

- In Sept 1996, the minimum wage was raised from $4.25 to $4.75
- Other US studies find that a 10% increase in the minimum wage increases teenage unemployment by 1 to 3%

Unemployment rate by education level in Switzerland



Source: SFSO Data represent the second quarter of each year

(4C) Unions and Collective Bargaining

- A ***union*** is a worker association that bargains with employers over wages, benefits and working conditions
- A union is a type of cartel attempting to exert its market power
  - Workers in a union act as a group when discussing their wages, benefits, and working conditions with their employers
- The process by which unions and firms agree on the terms of employment is called ***collective bargaining***
- If the union and firm do not reach agreement, the union can pressure employers to provide a higher wage level by threatening with strikes, i.e. the union organizes a withdrawal of labour from the firm
- By raising the wage level above equilibrium level (what they usually do), unions makes some workers better off and other workers worse off
  - Workers in unions (insiders) reap the benefits of collective bargaining, while workers not in the union (outsiders) bear some of the costs, e.g. unemployment. (***Insider-Outsider-Problem***)
- Empirical Evidence: Union workers earn 10 to 20 percent more than non-union workers

Are Unions Good or Bad for the Economy as a Whole?

- Economists disagree about whether unions are good or bad for the economy as a whole.
  - Critics argue that unions behave as cartels, cause the allocation of labour to be inefficient and inequitable:
    - Wages above the competitive level reduce the quantity of labour demanded and cause unemployment (inefficient)
    - Some workers benefit at the expense of other workers (inequitable)
  - Advocates of unions contend that unions are a necessary antidote to the market power of firms that hire workers. Example: Dominant employers
    - They claim that unions are important for helping firms respond efficiently to workers' concerns because unions represent workers' collective views on many job attributes and may help in finding the right mix of job attributes
- No consensus:

  Union's influence may be beneficial in some circumstances and adverse in others

(D) The Theory of Efficiency Wages

- ***Efficiency wages*** are above-equilibrium wages paid by firms in order to increase worker productivity.
- The theory of efficiency wages states that firms operate more efficiently if wages are above the equilibrium level. A firm may prefer higher than equilibrium wages for the following reasons:
  - Worker health:
    - Better paid workers eat a better diet, are healthier and are thus more productive.
  - Worker turnover:
    - A higher paid worker is less likely to look for another job. Firms save training costs, experienced worker are more productive than newly hired workers.
  - Worker quality:
    - Higher wages attract a better pool of workers to apply for jobs, because high-quality workers have higher reservation wages.
  - Worker effort:
    - Higher wages motivate workers to put forward their best effort.
- Real World Example: In 1914, Henry Ford's 5$ Workday (twice the prevailing wage of $2.34)

The Costs of Unemployment

- Costs for the individual
  - Loss of earnings
    - Increased risk of slipping into poverty
  - Self-esteem and health problems
  - Drug abuse and alcohol abuse and crime
  - Family breakdown
  - De-skilling
    - The individual loses touch with changes in work practices and technology

- Costs for society and the economy as a whole
  - The opportunity cost of unemployment
    - Lost output
  - The tax and benefits effect
    - Lower tax revenues,
      higher welfare payments
  - The Reverse Multiplier Effect:
    - People experiencing unemployment
      - Cut back their spending on luxuries
        - Causing these businesses to cut back orders, lay off workers, reduce profits
      - Switch their spending to substitute goods which may be seen as inferior goods
        - So, some firms might see demand increase e.g. budget supermarkets

Summary

- The unemployment rate is the percentage of those who would like to work but don't have jobs
- The unemployment rate is an imperfect measure of joblessness
- In the Swiss economy, most people who become unemployed find work within a short period of time
- Most unemployment observed at any given time is attributable to a few people who are unemployed for long periods of time
- One reason for unemployment is the time it takes for workers to search for jobs that best suit their tastes and skills
- A second reason why our economy always has some unemployment is minimum-wage laws
  - Minimum-wage laws raise the quantity of labour supplied and reduce the quantity demanded
- A third reason for unemployment is the market power of unions
- A fourth reason for unemployment is suggested by the theory of efficiency wages
  - High wages can improve worker health, lower worker turnover, increase worker effort, and raise worker quality

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Principles of Macroeconomics

Lecture 5: Saving, investment and the financial system

Prof. Dr. Jan-Egbert Sturm

Financial institutions

- **Finance** is the field of economics that studies how people make decisions regarding the allocation of resources over time and the handling of risk.
- The **financial system** is made up of institutions that coordinate the actions of savers and borrowers
- Financial institutions can be grouped into two different categories:
  - ***Financial markets*** are the institutions through which savers can directly provide funds to borrowers
  - ***Financial intermediaries*** are financial institutions through which savers can indirectly provide funds to borrowers

Financial institutions

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  - ***Financial markets*** are institutions through which savers can *directly* provide funds to borrowers
    - Bond Markets
      - A ***bond*** is a certificate of indebtedness that specifies obligations of the borrower to the bond holder. The associated credit risk is the probability that the borrower will fail to pay some of the interest or principal
      - ***Government bonds*** are often considered to be safe and therefore carry lower interest rates
    - Stock Markets
      - A ***stock*** represents a claim to partial ownership in a firm and is a claim to the profits that the firm makes
      - The issuing of stocks by a firm to raise money is called equity financing
      - Compared to bonds, stocks offer both higher risk and potentially higher returns
  - ***Financial intermediaries*** are institutions through which savers *indirectly* provide funds to borrowers

Financial institutions

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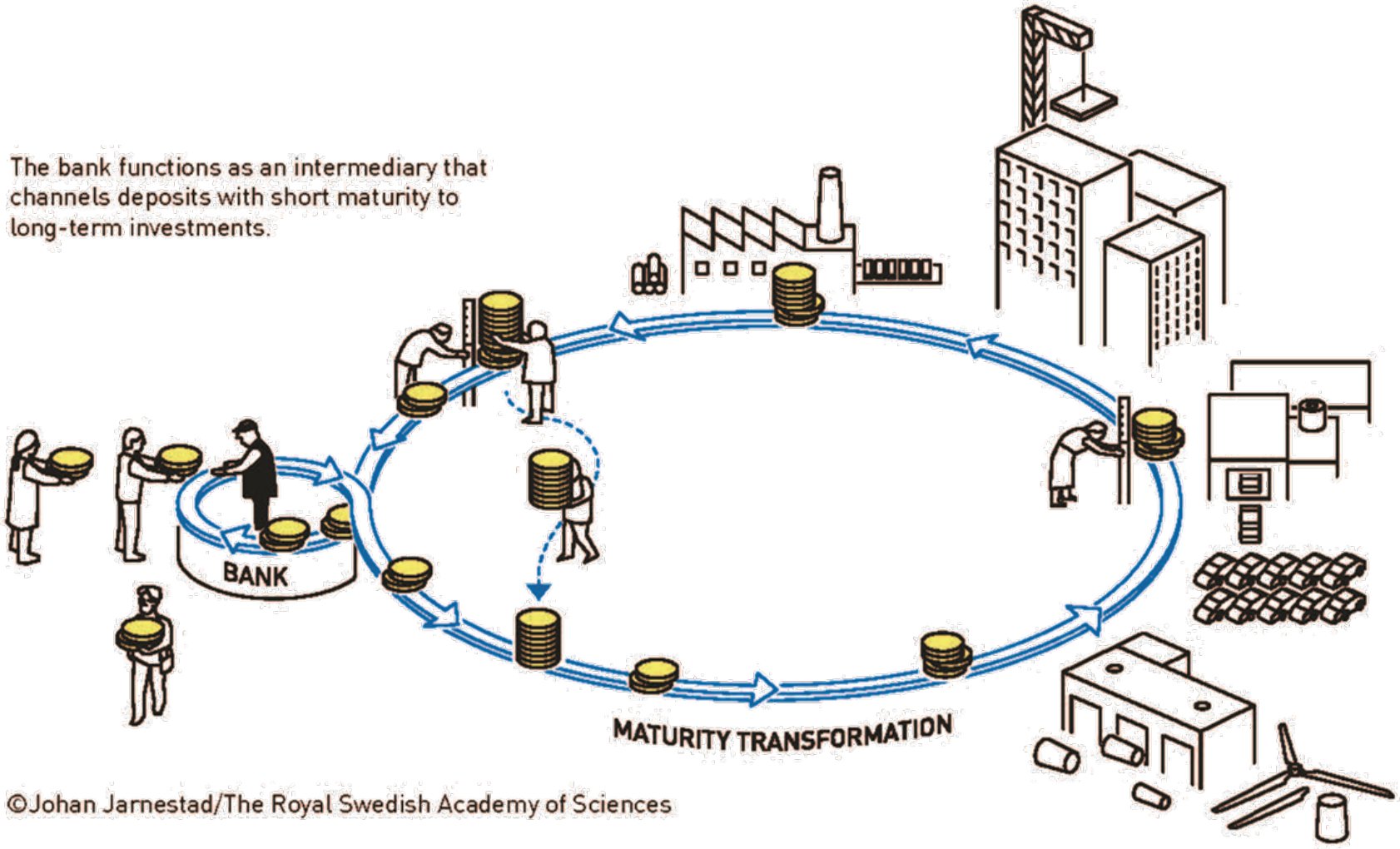
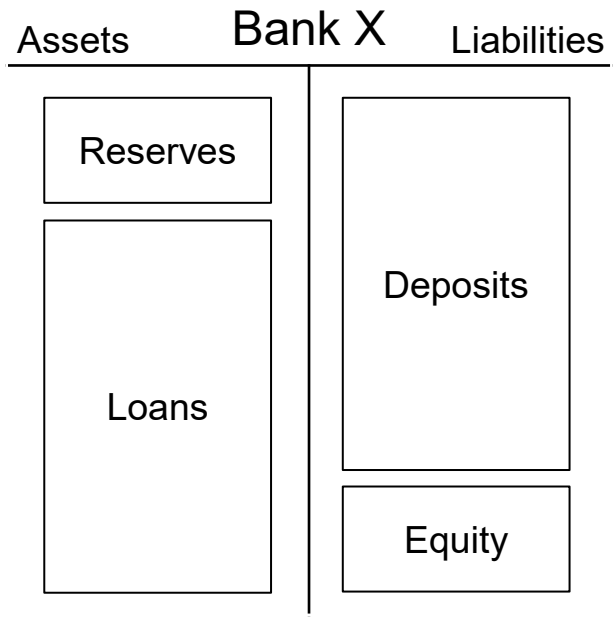
  - ***Financial markets*** are institutions through which savers can *directly* provide funds to borrowers
  - ***Financial intermediaries*** are institutions through which savers *indirectly* provide funds to borrowers
    - Banks
      - … take deposits from people who save and use these to lend to those who want to borrow
      - help create a medium of exchange by providing non-cash payment methods
    - Mutual Funds or Investment Funds
      - These funds are vehicles that allows the public to invest in a portfolio of various types of shares and/or bonds. These funds sell shares to the public and use the proceeds to buy such portfolios. They allow people with small amounts of money to easily diversify
    - Other Financial Institutions
      - Credit unions, Pension funds, Insurance companies, Loan sharks

The role of the financial sector



Banks and Macroprudential Policy



- Banks perform a maturity transformation.
  - Assets (such as loans) are typically long-term and illiquid.
  - Liabilities (such as deposits) are typically short-term and can be withdrawn on demand.
- Banks assume that not all depositors want to withdraw their deposits at once and keep only a small amount of cash/reserves ready (liquidity risk).
  - Banks cannot sell their assets quickly without a loss.
- If depositors lose trust in the solvency of the bank, they withdraw their deposits in cash.
  - This increases the likelihood of default and causes further withdrawals, triggering a bank run.

Banks and Macroprudential Policy

- The financial system is highly interconnected and interdependent. Failure of individual institutions may lead to contagion and cascades of defaults.
  - The risk of widespread failures across the entire financial sector is called systemic risk.

- Macroprudential policies are designed to limit risk and enhance stability in the financial sector.
  - Deposit insurance systems protect savings even if the bank fails.
  - Higher reserve and liquidity requirements.
  - Higher capital requirements.
  - Central banks act as lender of last resort.

Source: Finanz und Wirtschaft

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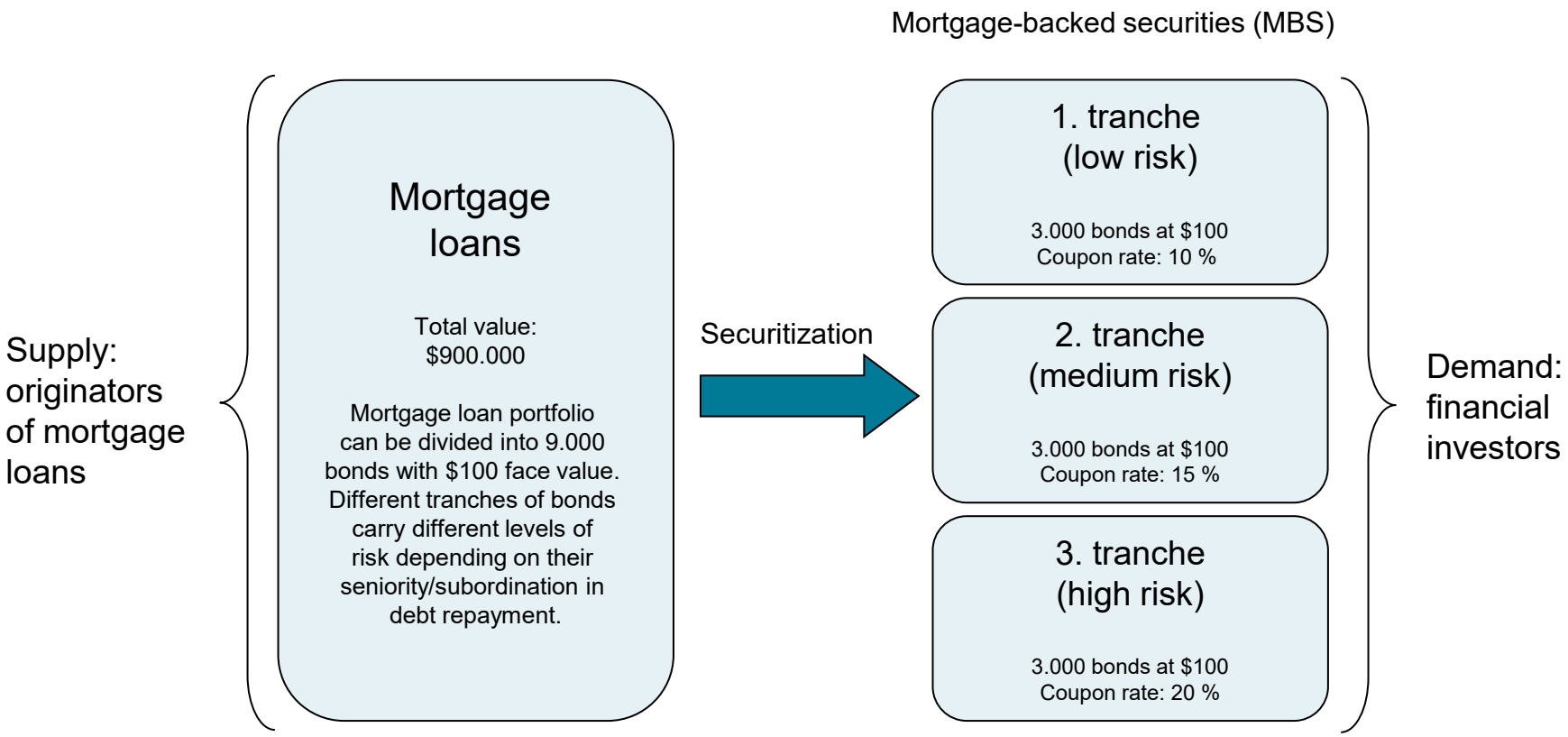
The Great Financial Crisis of 2007-2009

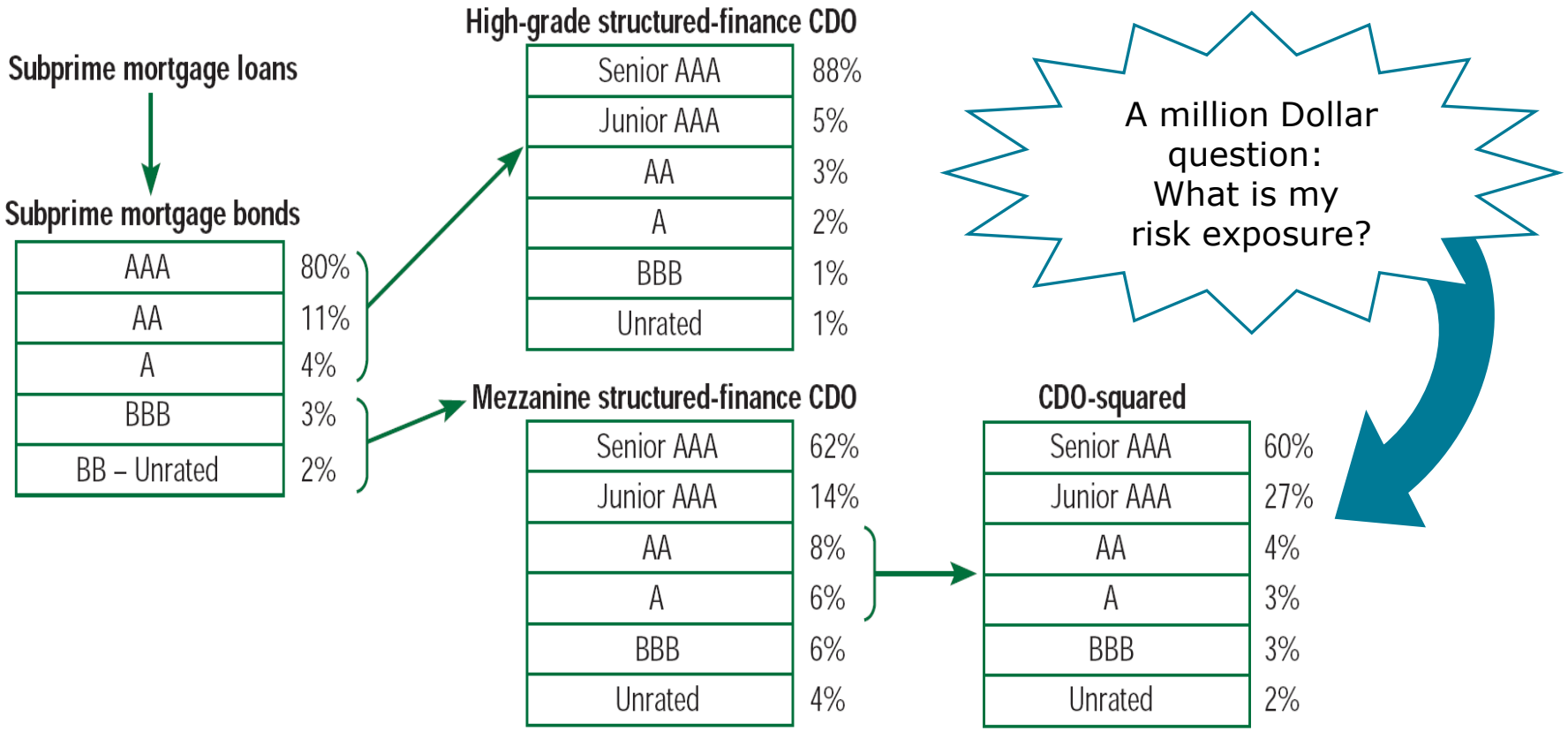
Structured Finance (1)

- Securitization involves packaging up loans (e.g. residential mortgages, car loans, credit card debt obligations) and selling them (typically via a special purpose vehicle) to investors.
- These financial products are referred to as collateralized debt obligations (CDOs).
  - They are pools of asset-backed securities which are dependent on..
    - .. the value of the assets that back them up.
    - .. the stream of income that flows from these assets (e.g. the interest paid on the mortgages).
- The issued securities are often split into tranches and rated according to their risk. Lower-risk tranches attract lower interest rates.
  - Credit rating agencies analyse the packages of debt and give them a credit rating.

Structured Finance (2)



Structured Finance (3)

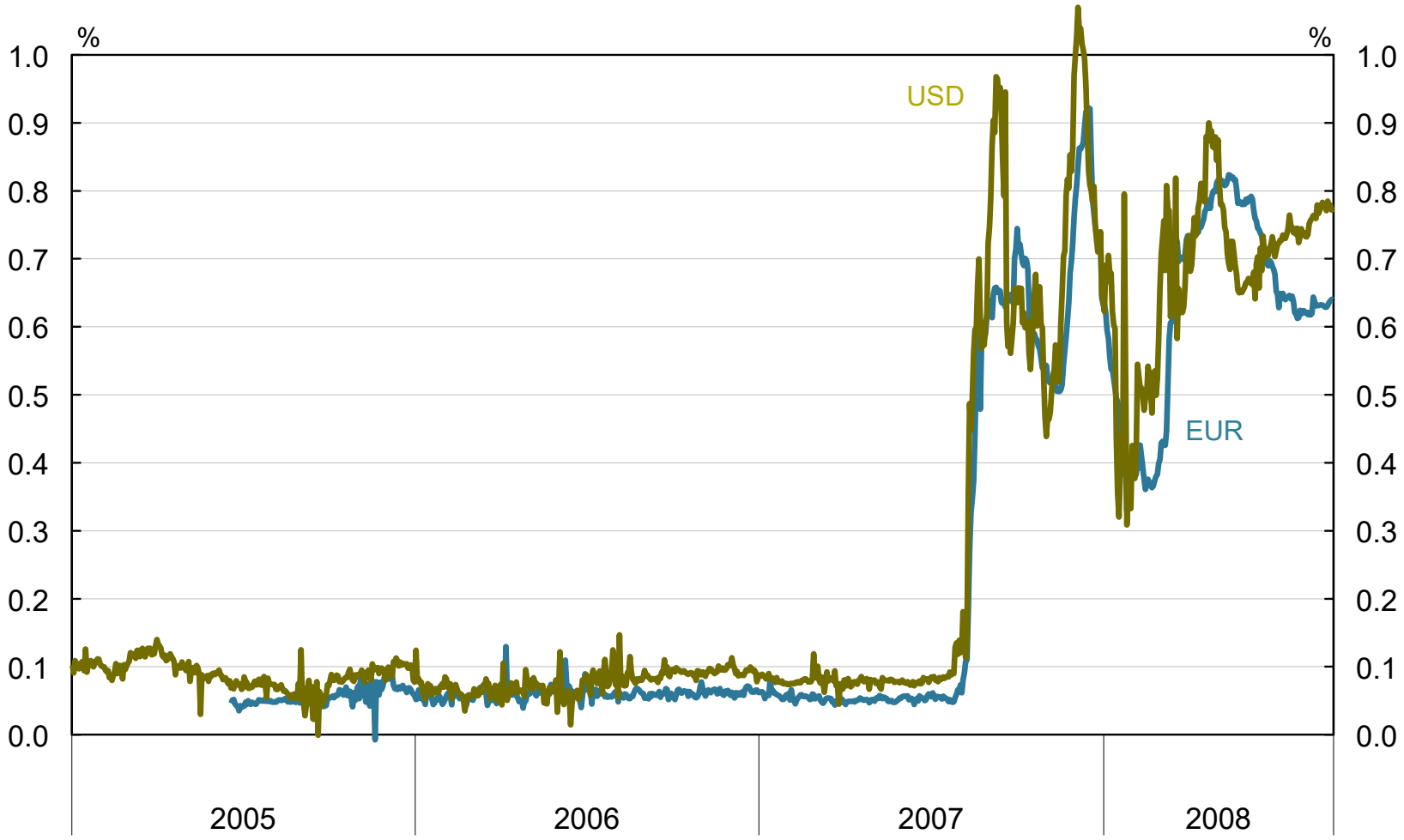


Source: IMF staff estimates.
Note: CDO = collateralized debt obligation.

Changes in Banking and Financial Innovation

- In the latter part of the 1990s and through to 2008, banks broadened the scope of their activities.
- Deregulation of financial markets in North America, Europe and the UK created the circumstances which allowed banks and other mortgage lenders to expand the amount of loans they made.
- Under ‘traditional’ lending regimes, banks would have had to have set aside reserves to cover the loans that they had made.
  - Securitization takes loans off the balance sheet, so the bank does not have to set aside reserves to cover those loans.
- In the run-up to the crisis, the market increasingly expanded from the prime market of safe lending and low risk into the sub-prime market, lending to individuals with high credit risks.
  - It was lending to people who had poor credit history.
  - Problems for CDOs began when many sub-prime mortgage holders began to default on payments.
  - CDOs became toxic assets because the value of the assets against which they were secured had fallen significantly.

Risk premium on the money market: Difference LIBOR and OIS (3-month rates)



Source: Datastream

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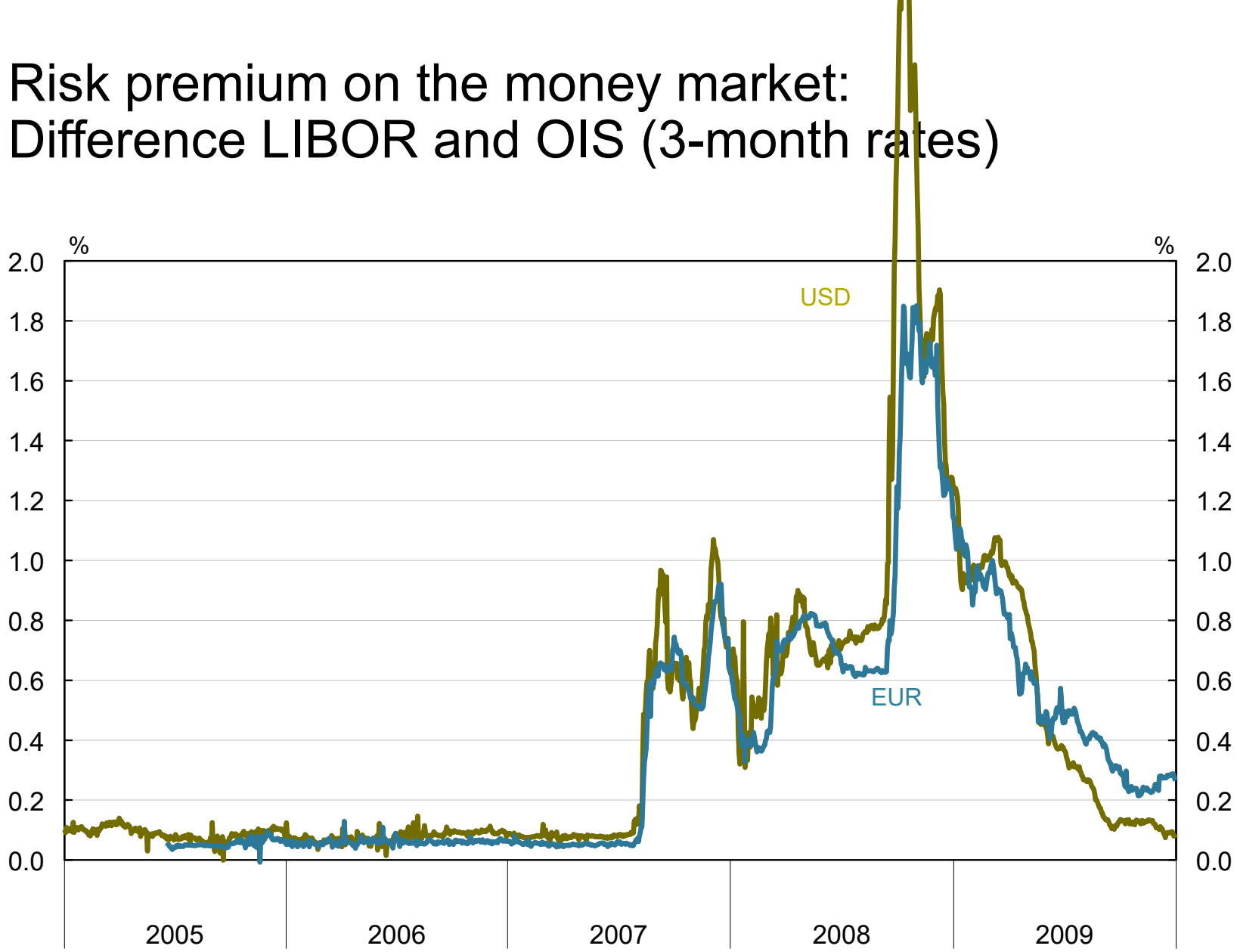
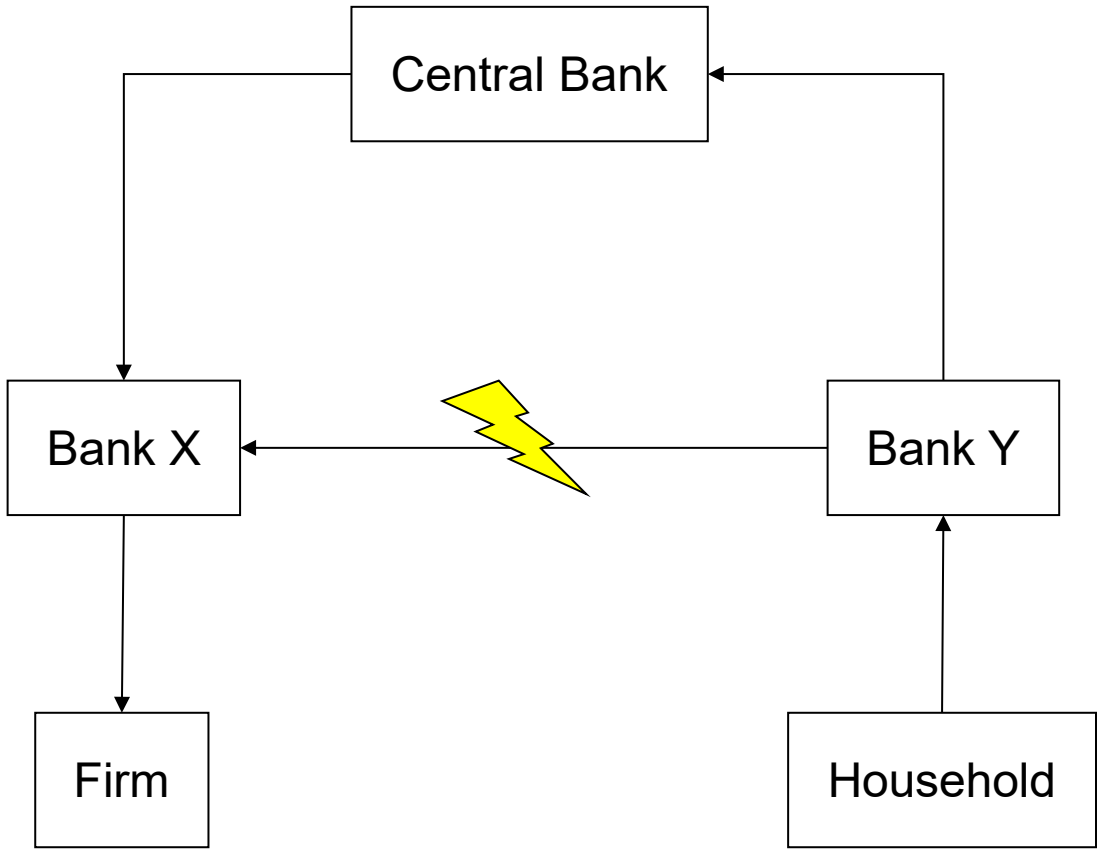


Source: Datastream

The Interbank Market



- Banks lend each other reserves on a short-term basis on the interbank market (money market).
- In 2008, it was hard to work out who would bear the losses resulting from the defaults in the subprime mortgage market.
  - Banks, as a result, became more cautious about lending to each other, leading to a credit crunch.
- Central banks had to step in and provide liquidity.

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Principles of Macroeconomics

Lecture 5: Saving, investment and the financial system

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Saving and Investment in the National Income Accounts

- Recall that GDP is both total income in an economy and total expenditure on the economy's output of goods and services:

$$Y = C + I + G + NX$$

- Assume a closed economy – one that does not engage in international trade (X=M=NX=0):

$$Y = C + I + G$$

- Now, subtract C and G from both sides of the equation:

$$Y – C – G = I$$

- The left side of the equation is the total income in the economy after paying for consumption and government purchases and is called national saving, or just saving (S).
  - Substituting S for Y – C – G, the equation can be written as: **S = I**
    - In macroeconomics, investment (I) refers to the purchase of new physical capital (and software)
    - If a person spends less than they earn and uses the rest either put in a bank, or to buy stocks, bonds, or investment funds, economists call this saving (S)

The Meaning of Saving and Investment

- ***National saving***, or saving, is equal to: $S = Y - C - G$
  $S = (Y - T - C) + (T - G)$ (where T stands for Taxes)
  - National saving is total income that remains after paying for consumption and government purchases
- ***Private Saving***
  - The part of income that households have left after paying taxes and paying for their consumption
  - Private saving = $(Y - T - C)$
- ***Public Saving***
  - Public saving is the amount of tax revenue that the government has left after paying for its spending
  - Public saving = $(T - G)$
  - Surplus and Deficit
    - If $T > G$, the government runs a ***budget surplus*** because it receives more money than it spends.
    - If $G > T$, the government runs a ***budget deficit*** because it spends more money than it receives
  - The accumulation of past budget deficits is called the government debt

Government Budget Balances (T-G) and Gross Government Debt in selected countries

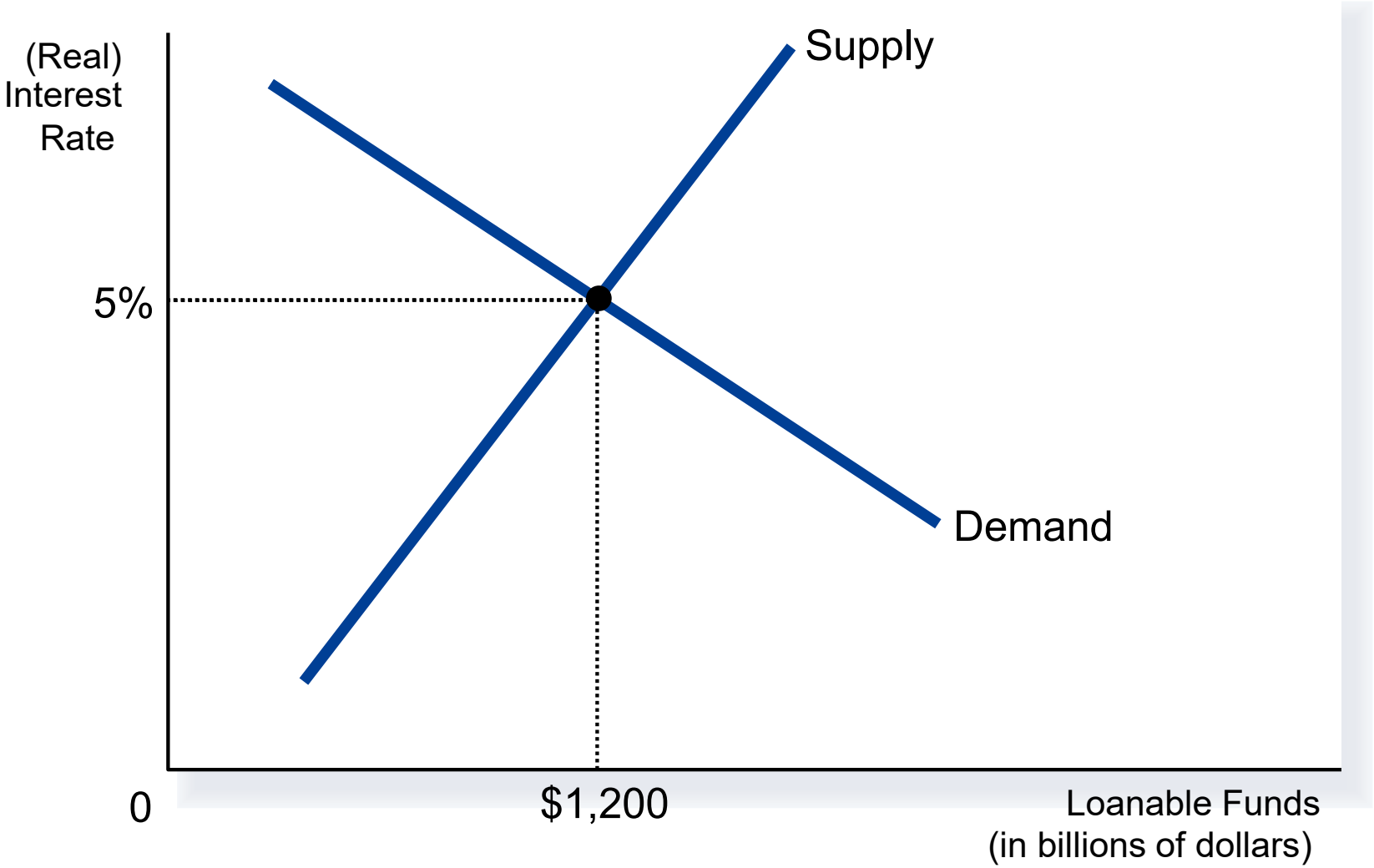
	Gross debt (as % of GDP)					Fiscal balance (as % of GDP)				
	2015–2021	2022	2023	2024	2025	2015–2021	2022	2023	2024	2025
Germany	65.6	65.0	62.9	62.5	63.8	-0.1	-2.1	-2.5	-2.8	-2.7
France	102.6	111.4	109.8	113.0	116.0	-4.5	-4.7	-5.4	-5.8	-5.6
Italy	138.7	138.3	134.6	135.3	136.7	-4.2	-8.1	-7.2	-3.4	-3.3
Spain	105.4	109.5	105.1	101.8	100.9	-5.0	-4.6	-3.5	-3.2	-2.8
Netherlands	54.8	48.4	45.2	43.3	45.0	-0.4	0.0	-0.4	-0.9	-2.1
Belgium	104.4	102.7	103.2	104.7	107.1	-3.3	-3.6	-4.1	-4.5	-5.4
Austria	79.9	78.4	78.5	81.8	84.0	-2.3	-3.4	-2.6	-4.7	-4.4
Euro area	**91.8**	**91.3**	**89.0**	**89.1**	**90.1**	**-2.5**	**-3.5**	**-3.5**	**-3.1**	**-3.2**
United States	111.9	120.8	122.7	124.1	125.4	-7.7	-3.7	-7.6	-7.5	-6.7
China	56.9	75.5	82.0	88.3	96.3	-5.0	-7.3	-6.7	-7.3	-8.6
Japan	239.0	256.9	250.1	247.8	246.4	-4.4	-4.2	-2.3	-2.5	-2.8
United Kingdom	92.2	99.6	100.4	101.3	102.6	-5.1	-4.6	-6.0	-6.0	-5.3
Switzerland	**41.2**	**37.2**	**38.7**	**37.6**	**36.9**	**0.2**	**1.3**	**0.2**	**0.1**	**-0.5**

Sources: European Commission; IMF World Economic Outlook

The Market for Loanable Funds

- The savings and investment rate determines the growth rate and long-run level of GDP
- Saving and investment are coordinated on the financial market
- We will assume for simplicity that the economy has only **one** financial market
  - This ***market for loanable funds*** is the market in which those who want to save supply funds and those who want to borrow to invest demand funds
  - Loanable funds refers to all income that people have chosen to save and lend out (S), rather than use for their own consumption
- The price for a loanable fund is the (real) interest rate
  - It reflects the amount borrowers pay for loans and lenders receive on their saving
- Financial markets link the present to the future and work much like other markets in the economy
  - A decision to save is a postponement of consumption in the present to consumption in the future
    - The intertemporal substitution effect is the rate at which economic actors respond to changes in the interest rate by changing savings and investment decisions
  - The equilibrium of the supply (S) and demand (I) for loanable funds determines the real interest rate

The Market for Loanable Funds - Graphical Illustration



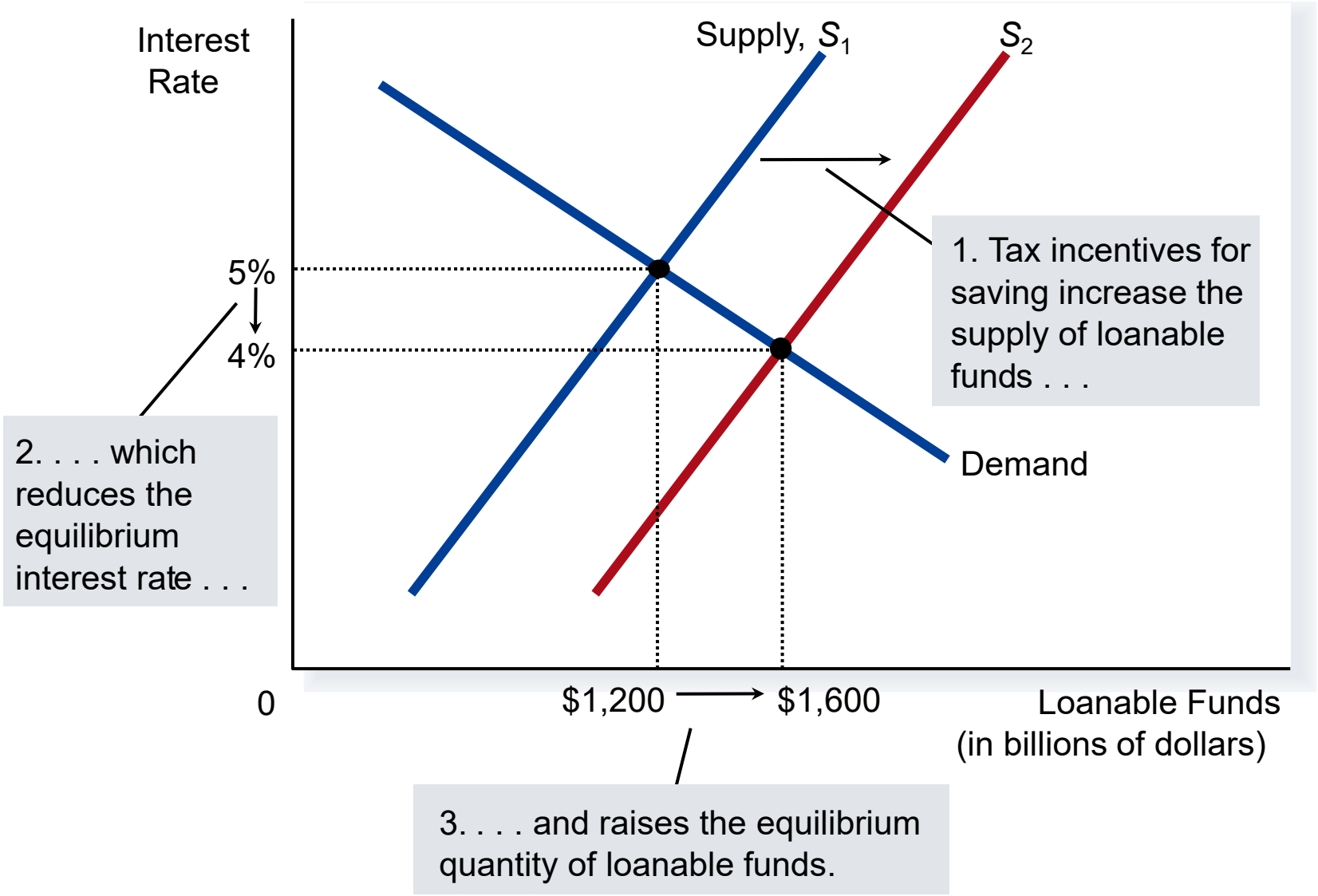
Government Policies to affect Savings and Investment

1. Taxes and saving
2. Taxes and investment
3. Government budget deficits and surpluses

Government Policies to affect Savings and Investment

1. Taxes and saving
   - Taxes on interest income substantially reduce the future payoff from current saving and, as a result, reduce the incentive to save
   - A tax decrease increases the incentive for households to save at any given interest rate
     - The supply of loanable funds curve shifts right
     - The equilibrium interest rate decreases
     - The equilibrium quantity of loanable funds increases
2. Taxes and investment
3. Government budget deficits and surpluses

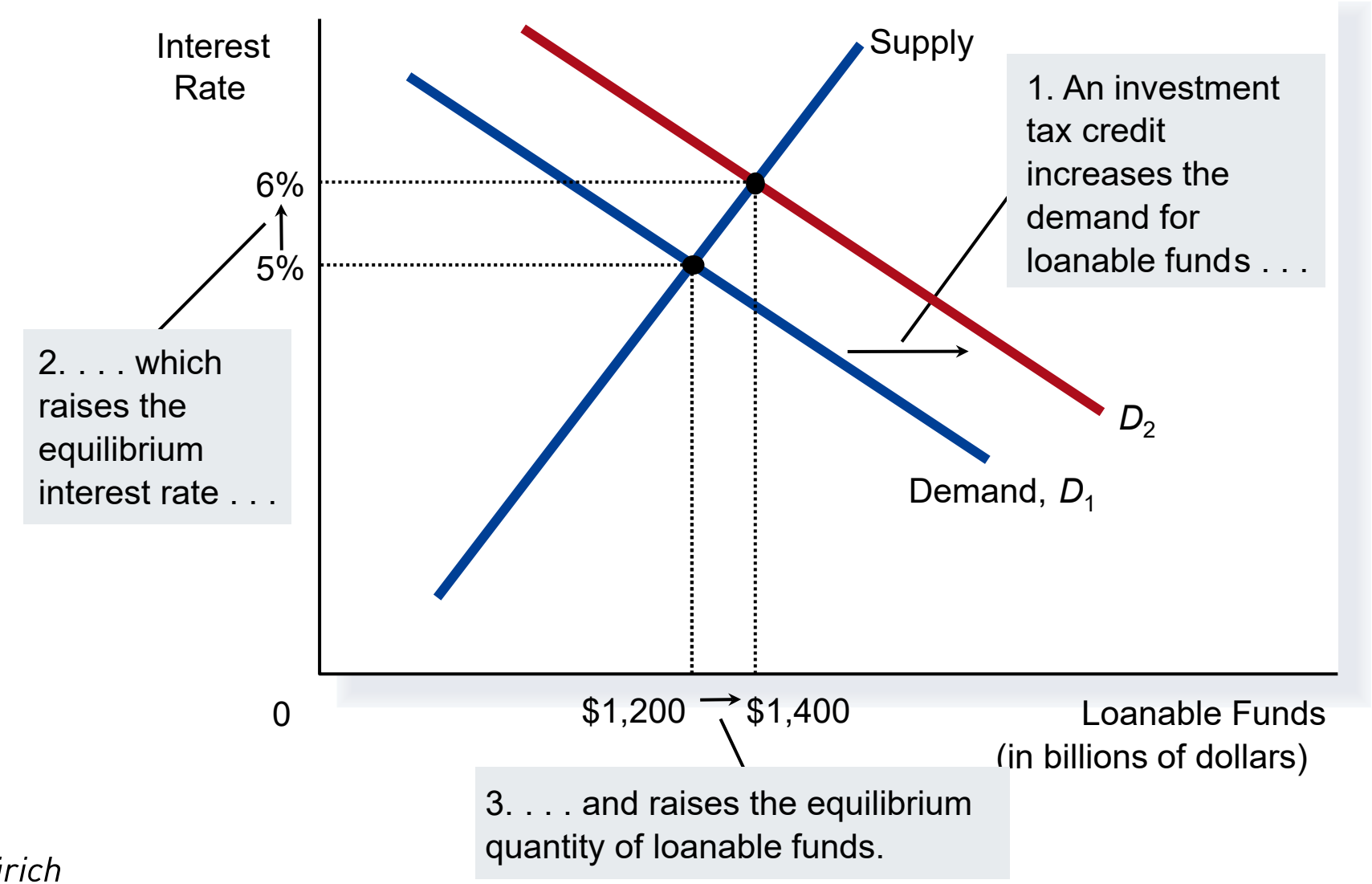
An Increase in the Supply of Loanable Funds



Government Policies to affect Savings and Investment

1. Taxes and saving
2. Taxes and investment
   - An investment tax credit
     - increases the incentive to borrow
     - increases the demand for loanable funds
     - shifts the demand curve to the right
     - results in a higher interest rate and a greater quantity saved
3. Government budget deficits and surpluses

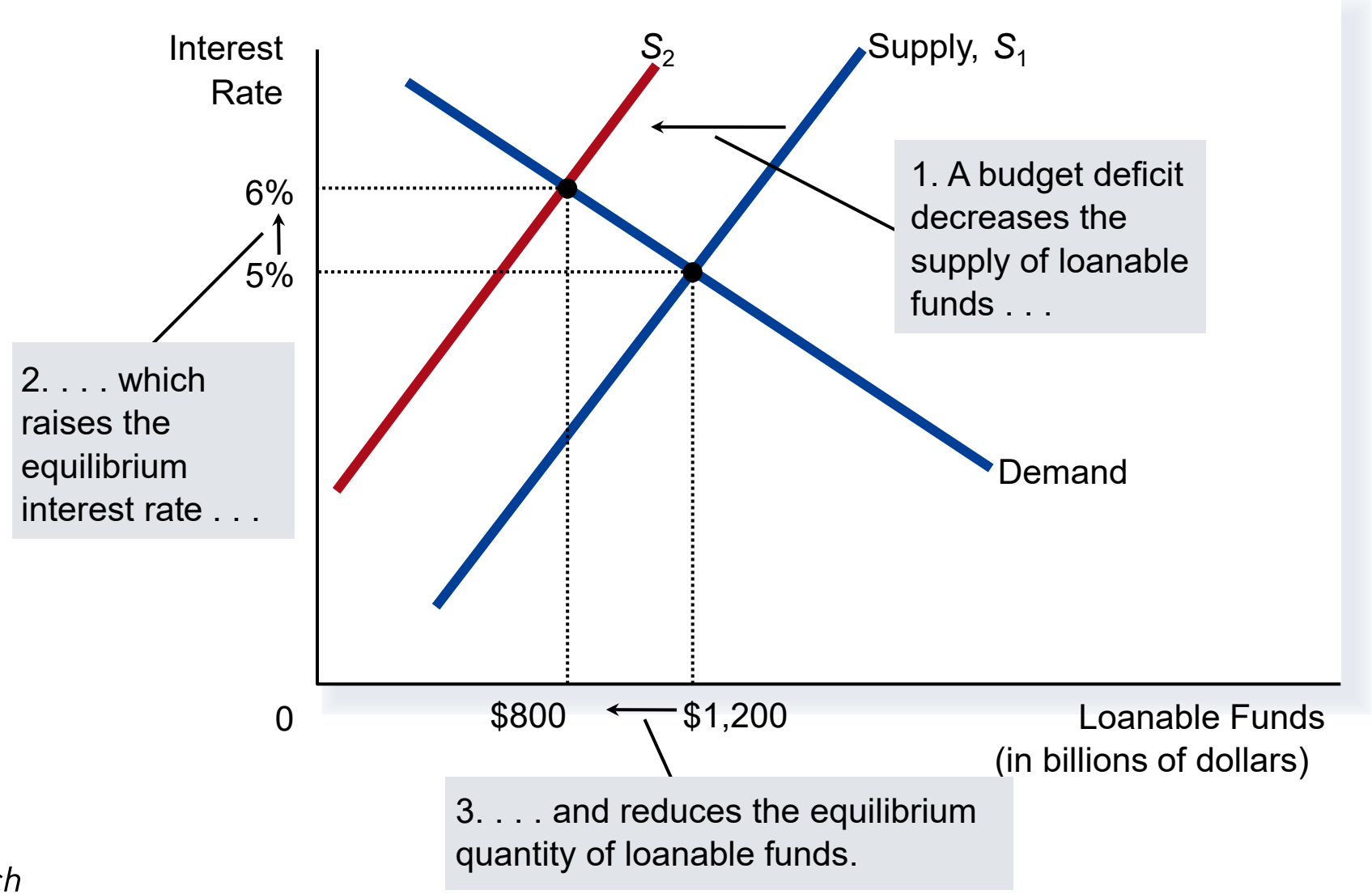
Investment Incentives Increase the Demand for Loanable Funds



Government Policies to affect Savings and Investment

1. Taxes and saving
2. Taxes and investment
3. Government budget deficits and surpluses
   - Government borrowing to finance its budget deficit reduces the supply of loanable funds available to finance investment by households and firms
   - A budget deficit decreases the supply of loanable funds
     - Shifts the supply curve to the left
   - Increases the equilibrium interest rate and reduces the equilibrium quantity of loanable funds
   - The associated fall in investment is referred to as **crowding out**
     - The deficit borrowing crowds out private borrowers who are trying to finance investments

The Effect of a Government Budget Deficit



Summary

- The financial system is made up of financial institutions such as the bond market, the stock market, banks, and mutual funds
- All these institutions act to direct the resources of households who want to save some of their income into the hands of households and firms who want to borrow
- National income accounting identities reveal some important relationships among macroeconomic variables
- In particular, in a closed economy, national saving must equal investment
- The savings and investment rate determines the growth rate and long-run level of GDP and consumption
- Financial institutions attempt to match one person's saving with another person's investment
- The interest rate is determined by the supply and demand for loanable funds
- The supply of loanable funds comes from households who want to save some of their income
- The demand for loanable funds comes from households and firms who want to borrow for investment
- National saving equals private saving plus public saving
- A government budget deficit represents negative public saving and, therefore, reduces national saving and the supply of loanable funds
- When a government budget deficit crowds out investment, it reduces the growth of productivity and GDP